

Why Can an AI Summary Be Factually Accurate and Still Be Misleading?

Context Loss, Semantic Compression, and Meaning Preservation

A. Jacobs | Reality Drift Framework

Abstract

A summary can contain only true sentences and still misrepresent its source. Meaning depends not only on isolated facts but on scope, sequence, uncertainty, attribution, contrast, and the relationship between claims. Compression changes those relationships. An AI system may retain the headline result while omitting the population, preserve a criticism while dropping the author's response, or combine observations into a conclusion the source never makes. This paper distinguishes factual accuracy from meaning preservation and explains why faithful summarization requires protecting constraints, not merely selecting correct propositions.

Source meaning → selection → compression → summary → reader interpretation

The Direct Answer

An AI summary can be factually accurate and misleading because truth at the sentence level does not guarantee fidelity at the document level. A summary changes emphasis, removes context, and creates new relationships among the details it retains.

The reader usually receives the summary as a model of the source. If the model preserves facts but changes the source's practical implication, the result is accurate in fragments and misleading as a whole.

Meaning Lives in Relationships

Documents communicate through more than propositions. They establish what is central, what is uncertain, which claim answers which objection, and which evidence applies under which conditions.

A summary that extracts correct statements but loses those relations can reverse argumentative weight. It may turn a qualified possibility into the apparent conclusion or make a minority position look representative.

Constraint Loss

Constraints specify where a claim holds. They include dates, populations, jurisdictions, definitions, comparison groups, confidence levels, and explicit exceptions.

Compression favors portable claims and treats constraints as secondary detail. Yet removing a constraint often changes the claim rather than merely shortening it. 'Associated in this sample' is not a longer version of 'causes.'

Selection Creates a Frame

Every summary chooses what deserves limited space. Selection creates a frame even when each selected fact is accurate. Repeated details appear important; omitted details disappear from the reader's model of the source.

AI systems often favor clear, answer-like material. Ambivalence, methodological limitation, and unresolved disagreement are less compressible, so they are especially vulnerable to omission.

Semantic Compression

Semantic Compression is the reduction of a larger meaning structure into a smaller representation. Good compression preserves the relationships necessary for the intended use. Poor compression preserves recognizable content while removing the constraints that governed interpretation.

This explains why a fluent summary can feel faithful during quick review. Familiar terms survive, but the reader may not notice that the source's structure has been rebuilt around the prompt.

Meaning Preservation

Meaning preservation can be evaluated by asking whether a reasonable reader would take the same central conclusion, confidence, scope, and practical implication from the summary as from the source.

Useful checks include preserving explicit uncertainty, labeling inference, retaining decisive counterevidence, and separating the source's claims from the summarizer's synthesis. Longer summaries are not automatically more faithful; the relevant issue is whether the right constraints survived.

Connection to Semantic Fidelity and Reality Drift

Semantic Fidelity names the preservation of meaning across transformation. It complements factual accuracy by examining relations, constraints, and interpretive consequences.

Within the Reality Drift Framework, misleading accuracy is especially dangerous because it remains operationally credible. The summary can be cited, circulated, and reused as if it were the source. Repeated reuse then hardens the compressed interpretation into the reality future systems encounter.

Conclusion

A summary is not faithful merely because every retained sentence is true. It must preserve the structure that determines what those sentences mean together. Accuracy protects the pieces. Meaning preservation protects the source's scope, uncertainty, relationships, and conclusion.